
Speaker script

“Back to entropy” — about 8:10 on the core path

Deck slides/talk.pdf — 17 slides + 5 backup slides
Version Final paper-aligned presentation and discussion guide
Target **under 10 minutes** of talking, then Q&A on the remaining time
Goal The team understands *how the method works* and *what we found*

How to use this document. Italic text is what to say — written to be spoken, not read aloud verbatim. Orange brackets are delivery cues. Times are cumulative: if you are past the stated time when you leave a slide, cut the next slide to its first sentence. The *must-land* points are marked **LAND THIS**; everything else is expendable under time pressure.

Length, measured not guessed. The spoken text is 1,205 words on the core path and 1,508 words in full, which at 150 wpm is **8:00** and 10:01 respectively. The cumulative times on each slide are the core path, which *omits slides 6, 7, 10, and 15* — the pipeline diagram, the contamination audit, and the math-domain bonus result. Those three are marked *optional* with the time they cost if you keep them; they are the detail/bonus slides that are the first thing to cut, not the must-land points. Re-check after any edit with `python tools/script_timing.py`, which checks that the core path stays under ten minutes.

The four things that must land:

1. Return to predictive entropy, made usable through byte normalization, fresh data, controlled adaptation, and contamination filtering.
2. **Adapted BPB selects the model**; adaptation moves candidates by as many as seven rank positions, while relative reduction explains corpus calibration.
3. BPB matches sentence continuation ($r = -0.975$, $\rho = 0.982$ vs. HellaSwag), while knowledge and arithmetic benchmarks add complementary capability views.
4. **The adapted ordering is domain-stable** (slide 14). All eleven models on three corpora: Spearman ρ between +0.955 and +0.991, Gemma-4-31B first on all three, and nothing moving more than two places. This is the cross-domain confirmation.

Talk shape. One ruler, one controlled intervention, three-domain evidence, one decision rule.

SLIDE 1

0:00–0:13

Title — “Back to entropy”

[Do not read the title. Land the one-sentence framing and move on.]

Quick version of what I have been building. The thesis: entropy, measured properly, gives us a direct and repeatable way to choose a base model for a target domain. Under ten minutes, then questions.

SLIDE 2

0:13–0:52

Return to the language-modeling objective

[LAND THIS] — point 1. *If you only get one slide across, it is this one.*

Language models were originally evaluated by predictive entropy. Then evaluation moved toward fixed task

suites: useful, but each one measures only a slice of base-model quality, and repeated optimization makes benchmark tuning and contamination difficult to rule out.

So we go full circle, but not back to raw perplexity. Perplexity is per token, and a token is a different amount of text in every model. Comparing it across tokenizers is like comparing euros and yen without an exchange rate. We make entropy useful again with four pieces: byte normalization, fresh target-domain text, controlled adaptation, and contamination filtering.

[The euros-to-yen line is the one people repeat afterwards. Deliver it slowly.]

SLIDE 3

0:52–1:50

Why not just read the scorecard?

[LAND THIS — this is the reason the method exists. Give the table three seconds of silence before you say the numbers. Do not rush to the punchline.]

The obvious question first. Every model ships with a scorecard — why measure anything ourselves?

Look at these two. On our ruler they are three per cent apart. On commonsense, two points apart. Essentially the same model, twice. On grade-school maths, one scores six per cent and the other sixty-one.

That gap is not telling you one model is better — the other two rows say they are tied. It is telling you what went into the two training mixes. Both are legitimate choices; the score just cannot tell you which one you are buying.

And note what I am not claiming. I am not saying anyone games their benchmarks; I cannot see anyone's training data. The point is structural. A years-old public benchmark is a target whether or not anybody aims at it, and text written after a model ships cannot have been studied for.

SLIDE 4

1:50–2:21

Measure the model we will actually use

[LAND THIS — point 2.]

A static benchmark describes the model as released. A product uses the model after it has seen some domain data. So we score both states on the same held-out corpus: zero-shot BPB, then adapted BPB.

The adapted level is the model-selection number because it measures the system we would actually deploy. The reduction between the two states has a different job: it tells us how much calibration the base distribution needed. One trajectory, two clearly defined outputs.

[Point to the right-hand box: that adapted model is the object being selected.]

SLIDE 5

2:21–3:04

Two measurements, one decision

[Do not derive the formula. Point at it, say what the denominator is, move on.]

Two measurements, one decision.

Move one: change the ruler. Normalize the loss by bytes instead of tokens — bits per byte. A UTF-8 byte is the same unit for every model on earth, so the numbers become comparable. BPB is not my invention — The Pile used it, Paloma uses it. It is just badly under-used, and we make it primary rather than a footnote.

Move two: measure domain fit. For each model, on one identical corpus: zero-shot BPB, a controlled LoRA adaptation, then adapted BPB. We rank by the adapted endpoint and keep the relative reduction as a calibration diagnostic. That gives every number one simple job.

The pipeline

[Walk the boxes left to right with your hand. Forty seconds, no more.]

Here is the whole thing. Raw corpus in; contamination audit; domain adaptation; BPB out. The ordering is strict by design: nothing reaches adaptation or scoring until it clears the contamination gate.

The corpus is a deduplicated slice of Google News from the eighth of June this year — about a hundred and twenty thousand articles — split with no overlap. Eleven base models, half a billion to thirty-five billion parameters: mostly dense Transformers, plus one sparse Mixture-of-Experts and one Liquid model, deliberately, to see whether the metric holds across genuinely different architectures.

Phase 1 — why we trust the numbers

[Technical slide. Give the shape of the idea, not the detail. If you are behind schedule, say only the first and last paragraphs.]

A word on the contamination audit, because it is what makes the headline result mean anything.

The honest headline first: for three of our four corpora we do not need a detector at all. Reddit, Hacker News and the math corpus were all scraped after every model in the cohort shipped, so they are clean by construction. Freshness is a stronger contamination defence than any screen, and it is the one we lean on.

The news corpus is the exception — June 2026, which overlaps part of the cohort's release window — so it gets the screen. Scout-then-screen: cheap detectors propose suspects, a perturbation test adds evidence. Two scouts: Min-K++, which flags odd confidence on a passage's rarest tokens, and CoDeC, which is ours — feed the model the text immediately preceding a passage and watch for a sharp jump in confidence. Then a quiz that hides the real passage among two perturbed variants.

Be careful how you say the last part: preference for the original is a risk signal, not proof of memorization — a model can prefer the fluent original having never seen it. We quarantine the union across all models, eighty sequences. No finite screen rules out pre-training exposure; only exact overlap is direct evidence.

Results — 11 models, one ruler

[The money slide. Read the axis once, then three models. Do not narrate all eleven. The exact table is in backup if anyone wants digits.]

Results. Eleven models, one ruler. One row per model, grey dot as released, blue dot after adaptation, further left is better; the label is the reduction.

Three readings. Gemma-4-31B ends furthest left at 0.624, the best adapted result. The thirty-five-billion Mixture-of-Experts lands with dense models in the twelve-to-fourteen-billion range, which gives us a useful quality measurement between its active and total parameter counts. And the Liquid model shows the largest calibration shift, 36.4 percent.

The endpoint chooses the model. The line length explains how much its base distribution moved toward the corpus. Keeping those roles separate makes the chart straightforward.

Two numbers, two jobs

[Keep this operational: endpoint for selection, reduction for explanation.]

The chart contains two useful numbers.

Adapted BPB is the primary selection score. It measures held-out target-domain prediction after the same kind of intervention we would use in deployment, and lower is better.

Relative reduction is the calibration diagnostic. It shows how much the released model adjusted to the corpus's style, formatting, and context pattern. It explains the path, while the endpoint makes the decision. We verified that interpretation token by token.

[Pause on "token by token" before advancing.]

Context explains the calibration signal

[LAND THIS — this is the new result and the answer to their question. Point at the crossing. Do not rush.]

We scored every token of the held-out set under the base and adapted models, then looked at where probability moved. Two patterns make the calibration interpretation concrete.

First, the picture. We aggregate corpus loss over independent five-hundred-and-twelve-token blocks, so almost every cut starts mid-article without the preceding context. The plot is byte-normalized, so tokenizer vocabulary size is no longer the simple explanation — and the curves still cross. Over the first eight positions Gemma averages 2.566 bits per byte against Qwen's 1.490; from position two-fifty-six on, Gemma is at 0.717 and Qwen at 0.773. With context, Gemma shows the stronger steady-state result. Adaptation quickly learns that context-recovery pattern.

Second, which tokens gain: full stops, commas, paragraph breaks, "the", "and", and "a". Within the top two hundred gain-carrying tokens, formatting, punctuation, and function words account for eighty-three percent of the mass. That is a clear signature of calibration to the corpus's surface form.

[Land the reporting rule: adapted level for selection, reduction for calibration.]

Adaptation does not improve the ranking — it rotates it

[LAND THIS — this is the central finding of the talk. Read the little table left to right, one row at a time. Do not rush it.]

Now compare both rankings — before and after adaptation — against the three benchmarks. The before column is free: it is one forward pass, no training.

And that free column is already good. Zero-shot news BPB agrees with GSM8K at 0.84 and MMLU-Pro at 0.86. After adaptation those fall to 0.73 and 0.76. HellaSwag goes the other way: 0.72 up to 0.98. The same three-way pattern repeats on Reddit and Hacker News.

So adaptation did not find a better model. It rotated the ranking — toward continuation quality and away from the generative tasks. And notice why HellaSwag agrees so well: it scores continuations by log-likelihood, which is the same object BPB measures. That is convergent validity, not proof the ranking improved.

Concretely: pick by zero-shot BPB and you get Qwen-3.5-35B, which is the actual GSM8K and MMLU leader. Pick by adapted BPB and you get Gemma-4-31B — the HellaSwag leader, and two to four accuracy points behind on the other two.

Different tasks reveal different strengths

[LAND THIS — the centrepiece. Slow down. Let people read the table for a beat before you talk over it.]

This is the table I find most useful in the whole project.

Eleven models. Left column is their BPB order; then each benchmark's own ordering of the same eleven, with a marker for how far each model moves.

The bottom two rows are the result. HellaSwag: seven of eleven in exactly the same position, and no model moves more than one place. MMLU-Pro, knowledge, drops to three of eleven. GSM8K, arithmetic, is zero of eleven — not one model keeps its place, and Gemma-4-12B moves five places.

The columns are complementary: BPB isolates target-domain language modelling, while the benchmarks add task-specific capability information. The trend is clear, but we do not claim task complexity forms a one-dimensional causal ladder.

[The 7/11, 3/11, 0/11 progression is the sentence to land before advancing.]

The family pattern explains the shift

[The two-model box on the right is the memorable example — point at it.]

The GSM8K shift follows a strong model-family pattern. Every Qwen is unchanged or moves up relative to its BPB position, while both Gemmas and Llama move down. This is consistent with different math/reasoning emphasis, although this observational comparison cannot isolate a causal training ingredient.

The box on the right is the sharpest version. Qwen-2.5-1.5B and Llama-3.2-1B are adjacent in BPB, and differ by only two points on HellaSwag. On GSM8K they score 0.611 and 0.065: a 54.6-point gap.

Same two models, two capabilities. The preferred model follows the capability the product needs.

Adaptation changes the ranking — then it transfers

[LAND THIS — adaptation changes the choice; the adapted result then transfers.]

Last result: all eleven models on news, Reddit, and Hacker News under one identical adaptation budget. First, adaptation materially changes the choice. Zero-shot versus adapted rank correlations are only 0.655, 0.609, and 0.682, with maximum moves of seven, seven, and five places. Gemma-4-31B moves from fourth, fifth, and sixth zero-shot to first adapted on news, Reddit, and Hacker News. So the adapted endpoint is not a rescaled zero-shot ranking.

Second, that adapted result transfers. Each panel compares two corpora; Pearson correlations are 0.992, 0.977, and 0.990, while Spearman ranges from 0.955 to 0.991. Gemma-4-31B is first on all three, and nothing moves more than two places across domains.

Every rank change sits inside a near-tied cluster. On Hacker News, Gemma-4-12B is only eight thousandths behind Qwen-MoE and nine behind Ministral. So adapted BPB identifies the winner and stable tiers; a target-domain run resolves the choice inside a near-tie.

[If asked about the MoE: three billion active parameters of thirty-five billion, and it scores like a dense 12–14B. BPB gives the direct quality measurement.]

Domain-matched adaptation narrows the GSM8K gap

[Cut this first under time pressure — it is a bonus result, not one of the four must-land points. If you keep it: this is the direct test of the open question from the GSM8K slide earlier.]

If the rotation is real, it should be steerable — so we steered it. Same protocol, but adapt on math-domain prose instead of news: arXiv abstracts, not GSM8K questions, screened against the GSM8K test set before any model saw it.

GSM8K agreement goes to 0.94, the best selector in the study — fifty of fifty-five model pairs ordered correctly — and the top pick goes back to being the actual GSM8K leader. It demotes exactly the Gemma models GSM8K already penalizes.

The controls are the important part. Reddit gives 0.69, Hacker News 0.79, news 0.73 — all at or below the 0.84 you get for free by not adapting at all. Only the topically matched corpus beats doing nothing. And it is not free either: HellaSwag agreement drops from 0.98 to 0.82.

If someone asks about significance: at eleven models the bootstrap interval on that difference still spans zero. The evidence is the pattern across four corpora, not any single contrast. Say that before they do.

We tried to break it. The ranking held.

Everything so far infers what adaptation does from how the ranking moves. So we tested it directly: we kept the adapters this time and scored the adapted models on the benchmarks.

The ordering came back identical in all twenty model-corpus-benchmark cells — rank correlation of one everywhere, not a single inverted pair. If adaptation were reshuffling models, that happens about once in a hundred and twenty times. So the claim we sell — that this ruler tells you which model is better — survived a direct attack at better than one percent.

We also re-scored the main study as head-to-head picks rather than correlations, taking the uncertainty over models. The metric picks the better model above chance in twenty-three of the twenty-four settings. And resampling the cohort twenty thousand times, matched adaptation steered the ranking the way we predicted in ninety-three and a half percent of draws, and the opposite way in one point four.

A simple decision rule

What we learned: BPB gives one coordinate system across tokenizers and architectures; adaptation does not improve the ranking so much as rotate it toward whatever the corpus exercises; that rotation is stable across three domains and holds through a 2,048-token inference context. So the adaptation corpus is not a detail of the setup — it is the question you are asking.

How to use it. Collect recent target-domain text; freshness beats any contamination detector. Then pick your tier. Free tier: rank by zero-shot BPB — one forward pass, and on three of our four corpora it picked the benchmark leader outright. Paid tier: adapt every candidate under one budget and rank by adapted BPB — worth it when you are actually going to fine-tune, or when one specific capability decides the choice. In that case match the corpus to the capability.

The bottom line: one fair ruler, two cost tiers, and a corpus you choose deliberately.

Thank you / questions

That is the tour. Happy to go deeper — there are backup slides with the benchmark scores, contamination numbers, training protocol, and longer-context result.

Discussion guide

Short answers to the most useful follow-up questions.

“What is BPB in one sentence?”

BPB is cross-entropy divided by UTF-8 bytes, so every tokenizer and architecture is measured in the same unit on the same text.

“What is new in this work?”

The contribution is the complete workflow: fresh domain data, contamination screening, controlled adaptation, adapted-BPB model selection, token-level interpretation, and three-domain validation.

“Why rank by adapted BPB?”

It measures the model after the same kind of lightweight domain intervention used in deployment. It is therefore the closest score to the system we would actually ship.

“What does the relative reduction tell us?”

It is a calibration diagnostic. The token analysis shows that formatting, function words, and context recovery carry most of the movement, so the reduction explains the path to the endpoint.

“Does 8.6% of Gemma’s gain come from the first 17 block positions?”

No. It is the injected document-start marker plus the next 16 tokens after each document boundary, which may occur anywhere inside a packed block. We call this the post-boundary window; the marker itself is masked from the final cross-domain BPB.

“Could Gemma’s vocabulary explain the large early loss?”

Vocabulary size can affect nats per token, so the final plot uses bits per UTF-8 byte. The crossing remains after byte normalization, showing that context-start sensitivity is the stronger explanation.

“Does the 512-token adapter generalize to longer contexts?”

Yes in our fixed-target check. Gemma’s base BPB continues improving from 0.6465 at 512 tokens to 0.6054 at 2,048, while adapted BPB stays essentially flat at 0.547. The adapter therefore retains a 9.6 percent advantage at four times its training context. Qwen’s smaller gain also remains positive.

“How does the contamination audit work?”

Min-K++ and CoDeC propose suspicious passages; the Dynamic Choice Quiz tests preference for the original over perturbations. We quarantine the union of risk-flagged passages across models so every candidate sees one identical screened corpus. This is conservative filtering, not proof of memorization.

“How do the static benchmarks complement BPB?”

HellaSwag confirms the sentence-continuation signal ($r = -0.975$, $\rho = 0.982$). MMLU-Pro and GSM8K add knowledge and arithmetic views; their rank agreements with BPB are 0.764 and 0.727.

“Why does the rank table use eleven models?”

The corrected analysis uses the complete cohort across BPB and all three static tasks, avoiding the stale nine-model subset that appeared in an earlier draft.

“How stable is the result across domains?”

Pairwise Spearman correlations range from +0.955 to +0.991 across news, Reddit, and Hacker News. Gemma-4-31B ranks first on all three, and no model moves more than two places.

“Does adaptation actually change which model we select?”

Yes. Zero-shot versus adapted rank correlations are only 0.655, 0.609, and 0.682 across the three corpora,

with maximum moves of seven, seven, and five places. Gemma-4-31B moves from ranks four, five, and six to first on all three after adaptation.

“What is the practical selection rule?”

Rank candidates by adapted BPB on recent target-domain text. Treat models within 0.01 BPB as a near-tied tier and use a domain-specific run to make the final choice.

“What does the MoE result mean?”

The 35B MoE has about 3B active parameters and scores alongside dense 12–14B models. BPB gives a direct quality measurement when neither active nor total parameter count is sufficient.

“What does this cost to run?”

The shared-budget comparison uses lightweight LoRA adaptation rather than full fine-tuning. It fits the existing Slurm allocation and produces one reusable score from one domain corpus.